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NIM: 103122400065

Kelas: SE0802

MODUL 3

TUGAS PENDAHULUAN

Aturan Tugas Pendahuluan :

1. Buatlah table menggunakan LiveSQL ataupun tools lainnya.
2. Screenshoot query dan hasil perubahan pada table.
3. Dilarang plagiarisme.

Soal Tugas Pedahuluan :

1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.
2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.
3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
 - a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)
 - b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d, [MONTH| YEAR]))
 - c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))
 - d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

1.

```
CREATE TABLE mahasiswa (  
    id NUMBER PRIMARY KEY,  
    nama VARCHAR2(100),  
    tempat_lahir VARCHAR2(50)  
    tanggal_lahir DATE,  
    nomor_hp VARCHAR2(15),  
    email VARCHAR2(50),  
    tinggi_badan NUMBER,  
    berat_badan NUMBER  
);
```

2.

```

INSERT ALL
INTO mahasiswa VALUES (1, 'Andi Wijaya', 'Jakarta', TO_DATE('2002-05-15', 'YYYY-MM-DD'), '081234567890', 'andi@mail.com', 170
INTO mahasiswa VALUES (2, 'Budi Santoso', 'Bandung', TO_DATE('2001-08-22', 'YYYY-MM-DD'), '081234567891', 'budi.s@mail.com',
INTO mahasiswa VALUES (3, 'Citra Lestari', 'Surabaya', TO_DATE('2003-01-10', 'YYYY-MM-DD'), '081234567892', 'citra_l@mail.com
INTO mahasiswa VALUES (4, 'Dedi Kurniawan', 'Medan', TO_DATE('2002-11-30', 'YYYY-MM-DD'), '081234567893', 'dedik@mail.com', 1
INTO mahasiswa VALUES (5, 'Eka Putri', 'Palembang', TO_DATE('2003-03-05', 'YYYY-MM-DD'), '081234567894', 'ekap@mail.com', 158
INTO mahasiswa VALUES (6, 'Fahmi Idris', 'Makassar', TO_DATE('2001-12-12', 'YYYY-MM-DD'), '081234567895', 'fahmi.i@mail.com',
INTO mahasiswa VALUES (7, 'Gita Gutawa', 'Yogyakarta', TO_DATE('2002-07-07', 'YYYY-MM-DD'), '081234567896', 'gita.g@mail.com'
INTO mahasiswa VALUES (8, 'Hadi Saputra', 'Semarang', TO_DATE('2001-04-18', 'YYYY-MM-DD'), '081234567897', 'hadis@mail.com',
INTO mahasiswa VALUES (9, 'Indah Permata', 'Denpasar', TO_DATE('2003-09-25', 'YYYY-MM-DD'), '081234567898', 'indah.p@mail.com
INTO mahasiswa VALUES (10, 'Joko Susilo', 'Solo', TO_DATE('2002-02-14', 'YYYY-MM-DD'), '081234567899', 'joko_s@mail.com', 178
INTO mahasiswa VALUES (11, 'Kartika Sari', 'Malang', TO_DATE('2003-06-20', 'YYYY-MM-DD'), '081234567800', 'kartika@mail.com',
INTO mahasiswa VALUES (12, 'Lutfi Hakim', 'Bogor', TO_DATE('2001-10-05', 'YYYY-MM-DD'), '081234567801', 'lutfi.h@mail.com', 1
INTO mahasiswa VALUES (13, 'Maya Ahmad', 'Aceh', TO_DATE('2002-01-25', 'YYYY-MM-DD'), '081234567802', 'maya_a@mail.com', 159,
INTO mahasiswa VALUES (14, 'Naufal Rizky', 'Padang', TO_DATE('2002-08-14', 'YYYY-MM-DD'), '081234567803', 'naufal@mail.com',
INTO mahasiswa VALUES (15, 'Olivia Rani', 'Manado', TO_DATE('2003-04-12', 'YYYY-MM-DD'), '081234567804', 'olivia@mail.com', 1
SELECT * FROM dual;

```

3.

a

```

SELECT
    LOWER(nama) AS nama_kecil,
    UPPER(nama) AS nama_besar,
    INITCAP(tempat_lahir) AS tempat_kapital,
    CONCAT(nama, email) AS gabung,
    LENGTH(nama) AS panjang_nama,
    SUBSTR(nama, 1, 3) AS tiga_huruf_awal,
    LPAD(id, 5, '0') AS id_pad,
    RPAD(nomor_hp, 15, '*') AS hp_sensor,
    LTRIM('  DATA') AS hapus_kiri,
    RTRIM('DATA  ') AS hapus_kanan,
    INSTR(email, '@') AS posisi_at,
    REPLACE(email, 'mail.com', 'gmail.com') AS ganti_domain
FROM mahasiswa;

```

	NAMA_KECIL	NAMA_BESAR	TEMPAT_KAPITAL	GABUNG	PANJANG_NAMA	TIGA_HURUF_AWAL	ID_PAD	HP_SENSOR	HAPUS_KIRI	HAPUS_KANAN
1	andi wijaya	ANDI WIJAYA	Jakarta	Andi Wijayaandi@mail.com	11	And	00001	081234567890***	DATA	DATA
2	budi santoso	BUDI SANTOSO	Bandung	Budi Santosobudi.s@mail.com	12	Bud	00002	081234567891***	DATA	DATA
3	citra lestari	CITRA LESTARI	Surabaya	Citra Lestarcitra_l@mail.com	13	Cit	00003	081234567892***	DATA	DATA
4	dedi kurniawan	DEDI KURNIAWAN	Medan	Dedi Kurniawandedik@mail.com	14	Ded	00004	081234567893***	DATA	DATA
5	eka putri	EKA PUTRI	Palembang	Eka Putriekap@mail.com	9	Eka	00005	081234567894***	DATA	DATA
6	fahmi idris	FAHMI IDRIS	Makassar	Fahmi Idrisfahmi.i@mail.com	11	Fah	00006	081234567895***	DATA	DATA
7	gita gutawa	GITA GUTAWA	Yogyakarta	Gita Gutawagita.g@mail.com	11	Git	00007	081234567896***	DATA	DATA
8	hadi saputra	HADI SAPUTRA	Semarang	Hadi Saputrahadis@mail.com	12	Had	00008	081234567897***	DATA	DATA
9	indah permata	INDAH PERMATA	Denpasar	Indah Permataindah.p@mail.com	13	Ind	00009	081234567898***	DATA	DATA
10	joko susilo	JOKO SUSILO	Solo	Joko Susilojoko_s@mail.com	11	Jok	00010	081234567899***	DATA	DATA
11	kartika sari	KARTIKA SARI	Malang	Kartika Sarikartika@mail.com	12	Kar	00011	081234567800***	DATA	DATA
12	lutfi hakim	LUTFI HAKIM	Bogor	Lutfi Hakimlutfi.h@mail.com	11	Lut	00012	081234567801***	DATA	DATA
13	maya ahmad	MAYA AHMAD	Aceh	Maya Ahmadmaya_a@mail.com	10	May	00013	081234567802***	DATA	DATA
14	naufal risky	NAUFAL RIZKY	Padang	Naufal Riskynaufal@mail.com	12	Nau	00014	081234567803***	DATA	DATA
15	olivia rani	OLIVIA RANI	Manado	Olivia Raniolivia@mail.com	11	Oli	00015	081234567804***	DATA	DATA

b	SELECT TO_CHAR(tanggal_lahir, 'MM/MON/MONTH') AS format_bulan, TO_CHAR(tanggal_lahir, 'DD/DY') AS format_hari, TO_CHAR(tanggal_lahir, 'YYYY/YY/RR') AS format_tahun, TO_CHAR(SYSDATE, 'HH24:MI:SS AM') AS waktu_skrng, ADD_MONTHS(tanggal_lahir, 12) AS tahun_depan, LAST_DAY(tanggal_lahir) AS akhir_bulan, NEXT_DAY(tanggal_lahir, 'MONDAY') AS senin_berikutnya, MONTHS_BETWEEN(SYSDATE, tanggal_lahir) AS usia_bulan, ROUND(tanggal_lahir, 'YEAR') AS pembulatan_tahun FROM mahasiswa;																																																																								
	<table><tr><th></th><th>FORMAT_BULAN</th><th>FORMAT_HARI</th><th>FORMAT_TAHUN</th><th>WAKTU_SKRNG</th><th>TAHUN_DEPAN</th><th>AKHIR_BULAN</th><th>SENI_BERIKUTNYA</th><th>USIA_BULAN</th></tr><tr><td>1</td><td>05/MAY/MAY</td><td>15/WED</td><td>2002/02/02</td><td>22:23:15 PM</td><td>15-05-2003</td><td>31-05-2002</td><td>20-05-2002</td><td>287,67525201612903225806451612903225806</td></tr><tr><td>2</td><td>08/AUG/AUGUST</td><td>22/WED</td><td>2001/01/01</td><td>22:23:15 PM</td><td>22-08-2002</td><td>31-08-2001</td><td>27-08-2001</td><td>296,44944556451612903225806451612903225806</td></tr><tr><td>3</td><td>01/JAN/JANUARY</td><td>10/FRI</td><td>2003/03/03</td><td>22:23:15 PM</td><td>10-01-2004</td><td>31-01-2003</td><td>13-01-2003</td><td>279,8365423387096774193548387096774193548</td></tr><tr><td>4</td><td>11/NOV/NOVEMBER</td><td>30/SAT</td><td>2002/02/02</td><td>22:23:15 PM</td><td>30-11-2003</td><td>30-11-2002</td><td>02-12-2002</td><td>281,191381048387096774193548387096774193548</td></tr><tr><td>5</td><td>03/MAR/MARCH</td><td>05/WED</td><td>2003/03/03</td><td>22:23:15 PM</td><td>05-03-2004</td><td>31-03-2003</td><td>10-03-2003</td><td>277,9978326612903225806451612903225806451612903225806</td></tr><tr><td>6</td><td>12/DEC/DECEMBER</td><td>12/WED</td><td>2001/01/01</td><td>22:23:15 PM</td><td>12-12-2002</td><td>31-12-2001</td><td>17-12-2001</td><td>292,7720262096774193548387096774193548387096774193548</td></tr><tr><td>7</td><td>07/JUL/JULY</td><td>07/SUN</td><td>2002/02/02</td><td>22:23:15 PM</td><td>07-07-2003</td><td>31-07-2002</td><td>08-07-2002</td><td>285,933316532258064516129032258064</td></tr></table>		FORMAT_BULAN	FORMAT_HARI	FORMAT_TAHUN	WAKTU_SKRNG	TAHUN_DEPAN	AKHIR_BULAN	SENI_BERIKUTNYA	USIA_BULAN	1	05/MAY/MAY	15/WED	2002/02/02	22:23:15 PM	15-05-2003	31-05-2002	20-05-2002	287,67525201612903225806451612903225806	2	08/AUG/AUGUST	22/WED	2001/01/01	22:23:15 PM	22-08-2002	31-08-2001	27-08-2001	296,44944556451612903225806451612903225806	3	01/JAN/JANUARY	10/FRI	2003/03/03	22:23:15 PM	10-01-2004	31-01-2003	13-01-2003	279,8365423387096774193548387096774193548	4	11/NOV/NOVEMBER	30/SAT	2002/02/02	22:23:15 PM	30-11-2003	30-11-2002	02-12-2002	281,191381048387096774193548387096774193548	5	03/MAR/MARCH	05/WED	2003/03/03	22:23:15 PM	05-03-2004	31-03-2003	10-03-2003	277,9978326612903225806451612903225806451612903225806	6	12/DEC/DECEMBER	12/WED	2001/01/01	22:23:15 PM	12-12-2002	31-12-2001	17-12-2001	292,7720262096774193548387096774193548387096774193548	7	07/JUL/JULY	07/SUN	2002/02/02	22:23:15 PM	07-07-2003	31-07-2002	08-07-2002	285,933316532258064516129032258064
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	KE_ANGKA	TGL_KE_TKS	TINGGI_KE_TKS	TEKS_KE_TGL
1	12345	15 May 2002	170 cm	19-03-2026
2	12345	22 August 2001	175 cm	19-03-2026
3	12345	10 January 2003	160 cm	19-03-2026
4	12345	30 November 2002	168 cm	19-03-2026
5	12345	05 March 2003	158 cm	19-03-2026
6	12345	12 December 2001	180 cm	19-03-2026
7	12345	07 July 2002	162 cm	19-03-2026
8	12345	18 April 2001	172 cm	19-03-2026
9	12345	25 September 2003	165 cm	19-03-2026
10	12345	14 February 2002	178 cm	19-03-2026
11	12345	20 June 2003	160 cm	19-03-2026
12	12345	05 October 2001	174 cm	19-03-2026
13	12345	25 January 2002	159 cm	19-03-2026
14	12345	14 August 2002	171 cm	19-03-2026
15	12345	12 April 2003	164 cm	19-03-2026

d

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SELECT
    id,
    tinggi_badan,
    ABS(-170) AS fungsi_abs,
    MOD(id, 3) AS fungsi_mod,
    FLOOR(tinggi_badan / 1.3) AS fungsi_floor,
    CEIL(tinggi_badan / 1.3) AS fungsi_ceil,
    POWER(id, 2) AS fungsi_power,
    ROUND(tinggi_badan / 3, 2) AS fungsi_round
FROM mahasiswa;

```

ID	TINGGI_BADAN	FUNGSI_ABS	FUNGSI_MOD	FUNGSI_FLOOR	FUNGSI_CEIL	FUNGSI_POWER	F
1	170	170	1	130	131	1	
2	175	170	2	134	135	4	
3	160	170	0	123	124	9	
4	168	170	1	129	130	16	
5	158	170	2	121	122	25	
6	180	170	0	138	139	36	
7	162	170	1	124	125	49	
8	172	170	2	132	133	64	
9	165	170	0	126	127	81	
10	178	170	1	136	137	100	
11	160	170	2	123	124	121	
12	174	170	0	133	134	144	
13	159	170	1	122	123	169	
14	171	170	2	131	132	196	
15	164	170	0	126	127	225	